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Finite element analysis of convection dominated reaction–diffusion problems

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Abstract

The numerical analysis of the CAU (*Consistent Approximate Upwind*) Petrov–Galerkin method of convection dominated reaction–diffusion problems is presented. The main issue in this analysis is that it considers elements with high interpolation orders and yields new definitions for the upwind functions and the local Peclet number in terms of the characteristic element h and the element interpolation order p. For regular solutions, the CAU method gets quasi-optimal convergence rate for the streamline derivative, keeping the same SUPG (*Streamline Upwind Petrov–Galerkin*) convergence rates. This improves the well-known h-version error analysis in [Comput. Methods Appl. Mech. Engrg. 45 (1984) 285] and the hp-version in [SIAM J. Numer. Anal. 37 (2000) 1618] for SUPG-like methods. It also improves the a priori analysis for shock-capturing methods presented in [Comput. Methods Appl. Mech. Engrg. 191 (2002) 2997].
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1. Introduction

It is well known that numerical simulations of convection-dominated problems present numerical difficulties related to the lack of stability: because convection dominates diffusion, classical Galerkin finite element methods generate unstable approximations, which usually exhibit spurious oscillations. The SUPG (*Streamline Upwind Petrov–Galerkin*) method, proposed by Brooks and Hughes [7], was the first variationally consistent, stable and accurate finite element model which had been successfully used to obtain approximate solutions of convection-dominated phenomena. For regular solutions this method presents quasi-optimal rate of convergence for the streamline derivative and was first analyzed by Johnson et al. [22]. This methodology has been combined with other techniques to solve convection

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dominated convection–diffusion space–time problems in many areas of interest [8,12,26]. Nevertheless, for nonregular solutions, localized oscillations are still observed in the neighborhood of steep gradients meaning that the streamline is not always the appropriate upwind direction. To overcome this lack of monotonicity many discontinuity capturing weighting terms were designed to enhance stability either in a linear [23,29] or in nonlinear [11,20,21] way. Despite the good results obtained with linear shock-capturing methods for piecewise linear or bi-linear finite elements, the extension to higher-order elements is less appropriate since the crosswind diffusion is added in a inconsistent way [24]. Quite promising results were obtained using the CAU (*Consistent Approximate Upwind*) finite element method proposed in [16]. This work introduced the idea that a better approximation for convection dominated problems can be obtained requiring the fulfillment of the original differential partial equation in each element for a modified velocity field that is chosen as closest to the real velocity field as possible in the L_2 sense. A quite similar approach was used later in [21] for time-dependent convection dominated flow problems. The systematic way of updating the upwind direction in the CAU method results in adding to the SUPG formulation a nonlinear discontinuity-capturing term in a consistent way, engendering an additional stability in the direction of the approximate gradient. As such control is introduced in a residual way, it has superior stability behavior when applied to transient and/or nonhomogeneous problems and when source term is present as pointed out by the authors [16]. Although this shock-capturing term was designed to vanish in regions of smooth solutions it was observed a loss of accuracy of the CAU approximation when the exact solution was regular. Improved accurate variants of CAU were presented later in [4,13]. These variants retain the SUPG accuracy as long as regular solutions are concerned and keep the same CAU stability in regions of sharp layers. However, the former is easily generalizable to multidimensional systems and this is the version used here. Besides, a remarkable improvement is observed in the approximation of boundary layers when this method is combined with a p or hp-refinement scheme as shown in [4,1]. The former work showed numerically that the extension to high-order elements requires new definitions for the stabilization parameters related to the SUPG and CAU operators in order to keep the standard error estimates of the SUPG method. This work showed that the local stabilization parameter, designed heuristically at that time, is $\mathrm{O}(h/p)$, independent of the solution regularity, where h is the characteristic element length and p is the element interpolation order. Using quadratic interpolation order, similar *ad hoc* design for the intrinsic time function of the SUPG method were proposed in [28,32], but all of them lack solid theoretical justification.

In [5,6], the hp-version of the discontinuous Galerkin FEM method for hyperbolic conservation laws was derived. They obtained the same Johnson's quasi-optimal error estimates for smooth solutions considering that the local stabilization parameter is $\mathrm{O}(h/p^2)$. However, the rate of the spectral convergence was suboptimal. In [18], the previous work was extended and the hp error analysis of the SUPG was derived, yielding that the stabilization parameter is indeed $\mathrm{O}(h/p)$, independent of the solution regularity. Similar result was obtained in [19] for the SUPG like method of partial differential equations with nonnegative characteristic form in the advection dominated limit. But, still, localized oscillations can appear in the vicinity of the sharp layers and, as shown in [4], they actually grow with the increase of the interpolation order.

As far as discontinuity capturing methods are concerned, the numerical analysis of nonlinear discontinuity-capturing finite element methods is not completely established yet. Some particular results can be found in [31,30]. In [24] one can find a very good contribution towards the existence of discrete solutions and some a priori and posteriori estimates. They showed that the shock-capturing schemes keep at least the accuracy of the basic SUPG scheme as far as certain structure of the shock capturing

term is preserved. It turns out that this condition is satisfied by the CAU method for linear interpolation. Following this analysis, we consider in the present paper the numerical analysis of the CAU method considering higher-order elements. We re-worked the SUPG error analysis for smooth solutions and high-order elements in Section 3. We derived the stability estimate with no dependence on the element interpolation order by re-defining the local stabilization parameter (or damping parameter/intrinsic time scale) and the local Peclet number. The effect of the nonlinear discontinuity-capturing stabilization introduced by the CAU is evaluated in Section 4, showing that this additional term does not degrade the optimal and quasi optimal SUPG convergence rates for regular solutions. We present the iterative solution of the nonlinear discrete problem and we highlight the role of the intrinsic time scale associated to the CAU operator in the accuracy and stability of the approximate solution. Some numerical experiments are performed in Section 5 and they verified the predicted estimates as far as regular solutions are concerned. Conclusions are drawn in Section 6.

2. Problem statement

Let $\Omega \subset \Re^d$ $(1 \leqslant d \leqslant 3)$ be a bounded domain with a Lipschitz continuous boundary Γ with an outward unit normal $\mathbf{n}$. We are interested in steady-state solutions of convection dominated reaction–diffusion scalar problems of the form

$$\mathcal{L}\phi := \gamma\phi + \mathbf{u}\cdot\nabla\phi - \varepsilon\triangle\phi = f \quad \text{in } \Omega, \tag{1}$$

$$\phi = g \quad \text{on } \Gamma, \tag{2}$$

with the following assumptions: $0 < \varepsilon \ll 1$; $\mathbf{u} \in W^{1,\infty}(\Omega)^d$; $\nabla\cdot\mathbf{u} = 0$, $\forall x \in \Omega$; $\gamma \in L^\infty(\Omega)$; $0 \leqslant \gamma \leqslant \sigma$; $g \in L^2(\Gamma)$ is a given function and $f \in L^2(\Omega)$ is a given source term. For simplicity, let us assume that γ and ε are constants. Thus, Eq. (1) represents the unknown field ϕ spreading in a medium characterized by the (small) diffusivity coefficient ε and is mainly transported by the velocity field $\mathbf{u}(\mathbf{x}) = (u_1, \ldots, u_d)$. Internal and external boundary layers can occur depending on g. Defining $V = H^1_g(\Omega)$ and $U = H^1_0(\Omega)$, a variational formulation for this problem reads: find $\phi \in V$, such that

$$a(\phi, \eta) = (f, \eta), \quad \forall \eta \in U, \tag{3}$$

with

$$a(\phi, \eta) = \varepsilon(\nabla\phi, \nabla\eta) + (\mathbf{u}\cdot\nabla\phi, \eta) + \gamma(\phi, \eta), \tag{4}$$

where $(\cdot,\cdot) = (\cdot,\cdot)|_\Omega$ denotes the $L^2(\Omega)$ inner product and $H^s(\Omega)$ denotes the usual Sobolev space of order s, with $H^1_t(\Omega) = \{\phi \in H^1(\Omega);\ \phi = t \text{ on } \Gamma\}$.

3. The SUPG method

Consider a finite quasi-uniform element partition consisting of N_e elements Ω_e such that $\overline{\Omega} = \bigcup_{e=1}^{N_e} \overline{\Omega}_e$ and $\bigcap_{e=1}^{N_e} \Omega_e = \emptyset$ (empty set). The characteristic element length is denoted by h. Let $V_h^p = V_h^p(h, p) \subset V$ (and $U_h^p \subset U$), $V_h^p = \{\eta_h \in C^0(\Omega);\ \eta_h|_{\Omega_e} \in P^p\ \forall \Omega_e,\ \eta_h|_\Gamma = g\}$, where P^p is the set of interpolation polynomials of degree less or equal to p defined in each element Ω_e. The SUPG method [7] consists of find $\phi_h \in V_h^p$, such that

$$D(\phi_h, \eta_h) = F(\eta_h), \quad \forall \eta_h \in U_h^p, \tag{5}$$

with

$$D(\phi_h, \eta_h) = a(\phi_h, \eta_h) + \sum_{e=1}^{N_e} (\mathcal{L}\phi_h, \tau_s \mathbf{u} \cdot \nabla \eta_h)|_{\Omega_e}, \tag{6}$$

$$F(\eta_h) = (f, \eta_h) + \sum_{e=1}^{N_e} (f, \tau_s \mathbf{u} \cdot \nabla \eta_h)|_{\Omega_e}. \tag{7}$$

For $\phi \in H^2(\Omega) \cap H_g^1(\Omega)$ we have the consistency of the method:

$$D(e, \eta_h) = 0, \quad \forall \eta_h \in U_h^p,\ e = \phi - \phi_h. \tag{8}$$

The stability property of the bilinear form $D(\cdot, \cdot)$ is

$$D(\eta_h, \eta_h) \geqslant C(\theta) |\!|\!| \eta_h |\!|\!|^2, \quad \forall \eta_h \in U_h^p, \tag{9}$$

where $C(\theta) = 1 - (\sqrt{1+\theta})/2$ and the *stabilized energy mesh dependent norm* is

$$|\!|\!| \eta_h |\!|\!|^2 := \varepsilon \|\nabla \eta_h\|^2 + \gamma \|\eta_h\|^2 + \sum_{e=1}^{Ne} \left\{ \tau_s \|\mathbf{u} \cdot \nabla \eta_h\|_{\Omega_e}^2 \right\}. \tag{10}$$

The stability result (9) and the continuity of (6) yield uniqueness and existence of (5) and it is obtained assuming the following minimal design condition over the SUPG stabilization parameter τ_s [17]:

$$\exists \theta \in (0, 3) \quad \text{such that } \varepsilon \tau_s C_{\text{inv}}^2 \leqslant \frac{\theta}{2} h^2. \tag{11}$$

This condition has extended the criteria introduced in [22] in order to guarantee that the higher order term of the SUPG contribution $(-\varepsilon \triangle \eta_h, \tau_s \mathbf{u} \cdot \nabla \eta_h)|_{\Omega_e}$ does not degrade the extra stability introduced by the term $(\mathbf{u} \cdot \nabla \eta_h, \tau_s \mathbf{u} \cdot \nabla \eta_h)|_{\Omega_e} = (\tau_s \mathbf{u} \otimes \mathbf{u} \nabla \eta_h, \nabla \eta_h)|_{\Omega_e}$. The inverse estimate constant C_{inv} (with $C_{\text{inv}} = 0$ if $p = 1$) is usually given by $\|\triangle \eta_h\|_{\Omega_e} \leqslant C_{\text{inv}} h^{-1} \|\nabla \eta_h\|_{\Omega_e}$ (cf. [10]). If the dependence on p is gotten from C_{inv}, the inverse estimate is written as presented in [27],

$$\|\triangle \eta_h\|_{\Omega_e} \leqslant \overline{C}_{\text{inv}} p^2 h^{-1} \|\nabla \eta_h\|_{\Omega_e}, \tag{12}$$

such that $C_{\text{inv}} = \overline{C}_{\text{inv}} p^2$, where $\overline{C}_{\text{inv}}$ denotes a constant which does not depend on h and p. Thus, the condition (11) is primarily a stability requirement at the diffusive limit and, in some extend, requires that the SUPG contribution $\sum_{e=1}^{N_e} (\mathcal{L}\phi_h, \tau_s \mathbf{u} \cdot \nabla \eta_h)|_{\Omega_e}$ vanishes as fast as the diffusion term becomes dominant. This has led to the commonly used τ_s definition, independent of p, for a squared $(h \times h)$ regular mesh aligned with the $\mathbf{u}$ direction [7]:

$$\tau_s = \frac{h}{2|\mathbf{u}|} \xi(P_e), \tag{13}$$

where $\xi(P_e)$ is the dimensionless function that depends explicitly on the local Peclet number ($P_e = \frac{|\mathbf{u}|h}{2\varepsilon}$). Generalization of the definition (13) to unstructured meshes can be done as in [7].

The optimal ξ function (optimal in the sense that it leads to nodally exact solution for the one-dimensional steady convection diffusion problem with $f = 0$) is given by $\xi(P_e) = \coth P_e - 1/P_e$. An alternative definition, the critical approximation $\xi(P_e) = \max\{0, 1 - 1/P_e\}$ [9], can also be used to improve computational efficiency. This latter choice was justified by the Fourier analysis presented recently in [14] with grid-aligned and linear/bi-linear elements. The local Peclet number plays the role of

controlling the transition between the local diffusive dominated towards the local convective dominated behaviors. The asymptotic trend of $\xi(P_e)$ as P_e increases keeps the stability at the convective dominated limit ($\xi(P_e) = 1$) and the slope of $\xi(P_e)$ at the origin attains the stability at the diffusive limit ($\xi(P_e) = 0$). The choice (13) satisfies the previous requirement (11) for $p = 1$ (or $C_{\text{inv}} = 0$), or, in other words, $\xi(P_e) \leqslant \frac{2\theta}{C_{\text{inv}}^2} P_e$, imposing a stability region for $\xi(P_e)$. For higher-order elements, C_{inv} increases and $\xi(P_e)$ must not grow more than linearly with P_e, with the slope $2\theta/C_{\text{inv}}^2$. In order to take into account the degree of the interpolation keeping the definition (13), it was suggested in [15] to perform a re-scale of the usual element Peclet number, dividing it by C_{inv}^2. Doing so would preclude the need of changing the requirements for deriving error estimates. We advocate here a new design for both, the Peclet number and the τ_s parameter, still keeping the only requirement (11). First of all notice, that for a given ratio $|\mathbf{u}|/\varepsilon$, the local Peclet number must decrease either as h decreases for a given p ($P_e \to 0$ as $h \to 0$) or as the increase of the interpolation order for a given h ($P_e \to 0$ as $p \to \infty$). Moreover, the upwind parameter τ_s should be reduced either as p increases for a given h ($\tau_s \to 0$ as $p \to \infty$) or as h decreases for a given p ($\tau_s \to 0$ as $h \to 0$). This indicates that both τ_s and P_e must incorporate the effect of the degree of the interpolation order. These remarks amount to selecting

$$\tau_s^p = \frac{h}{2|\mathbf{u}|p}\xi\left(P_e^p\right) \quad \text{and} \quad P_e^p = \frac{|\mathbf{u}|h}{2\varepsilon p}. \tag{14}$$

Note that these definitions still satisfy the requirement (11). These choices are also in agreement with the heuristic designs used in [28,32] for quadratic interpolation order. More recently, the error estimates of the SUPG method was performed in [18] for first-order hyperbolic problems: assuming that $\phi|_{\Omega_e} \in H^{k+1}(\Omega_e)$ for some $k \geqslant 1$, for any $0 \leqslant s \leqslant \min(p, k)$, $p \geqslant 1$ and, in particular, if s is fixed and $p \gg 1$, the authors obtained the following estimates whenever the stabilization parameter is $\mathrm{O}(h/p)$:

$$|||\phi - \phi_h|||^2 \leqslant C \sum_{e=1}^{N_e} \left(\frac{h}{p}\right)^{2s+1} |\phi|_{s+1,\Omega_e}^2, \tag{15}$$

where the constant $C > 0$ depends only on elemental shape regularity. The previous estimate is not very useful if the solution does not have the required regularity. However, it shows that the choice (14) is appropriate at the hyperbolic limit in order to keep the quasioptimal estimates in both h and p for regular solutions.

The generalisation of the a priori error estimate for the SUPG like method of the purely hyperbolic problem and the elliptic advection–diffusion problem with the diffusivity coefficient ε isotropic and constant is presented in [19]. Similar global error bound was obtained in the advection dominated limit whenever the stabilization parameter is $\mathrm{O}(h/p)$. The reader should look to [18,19] for more general situations.

4. The CAU method

The SUPG method does not have the required stability to control derivatives in the gradient direction when high gradients characterized by internal and/or external boundary layers occur, resulting in local oscillations in the neighborhood of sharp gradients. They can even be amplified if high-order interpolation elements are used in these regions [4]. To improve stability the CAU (Consistent

Approximate Upwind) formulation was introduced in [16] for the scalar problem. Its generalization to the system of compressible Navier–Stokes equations was performed later in [2,3], where it was shown its feature of precluding oscillations in the presence of shocks and boundary layers.

The CAU method is a stabilized finite element method that consistently introduces, besides the SUPG contribution, a nonlinear discontinuity-capturing term that controls the derivatives in the direction of the approximate gradient. For the sake of clarity, the main features of this method is presented in the following.

Within the same finite element space V_h^p the CAU formulation for the problem (1) is: find $\phi_h \in V_h^p$ such that

$$D(\phi_h, \eta_h) + \sum_{e=1}^{N_e} \big(\mathcal{L}(\phi_h) - f, \tau_c[\mathbf{u} - \mathbf{v}] \cdot \nabla \eta_h\big)|_{\Omega_e} = F(\eta_h), \quad \forall \eta_h \in U_h^p, \tag{16}$$

where τ_c is another stabilization parameter, to be defined later, and the SUPG operator is defined from now on using τ_s^p. The auxiliary vector field $\mathbf{v}$ is a modified velocity field chosen such that it satisfies the original differential partial equation (1) for the approximate solution $\phi_h \in V_h^p$ in each element. This requirement implies

$$\big(\mathbf{u} - \mathbf{v}(\phi_h)\big) \cdot \nabla \phi_h = \mathcal{L}(\phi_h) - f = R(\phi_h). \tag{17}$$

The vector field $\mathbf{v}$ is determined such that it is the velocity field closest to the real velocity field $\mathbf{u}$ in the L_2 sense. Hence, it is obtained by solving the following local minimization problem: find $\mathbf{v} \in L^\infty(\Omega)^d$ such that

$$\|\mathbf{u} - \mathbf{v}\|_{\Omega_e} \leqslant \|\mathbf{u} - \nu\|_{\Omega_e}, \quad \forall \nu \in Q_\nu, \tag{18}$$

with

$$Q_\nu = \{\nu;\ \nu \cdot \nabla \phi_h - \varepsilon \Delta \phi_h + \gamma \phi_h - f = 0 \text{ in } \Omega_e,\ e = 1, 2, \ldots, N_e\}.$$

The solution of (18) together with (17) leads to

$$\begin{cases} \mathbf{v}(\phi_h) = \mathbf{u} & \text{if } |\nabla \phi_h| = 0, \\ \mathbf{u} - \mathbf{v}(\phi_h) = \frac{R(\phi_h)}{|\nabla \phi_h|^2} \nabla \phi_h & \text{if } |\nabla \phi_h| \neq 0. \end{cases} \tag{19}$$

Thus, if $|\nabla \phi_h| \neq 0$ we obtain $|\mathbf{u} - \mathbf{v}| = |R(\phi_h)|/|\nabla \phi_h|$, which ensures that, in each element Ω_e, $\lim_{h \to 0} \mathbf{v}(\phi_h) = \mathbf{u}$ when $\lim_{h \to 0} \phi_h = \phi$. Besides, as $\mathbf{v}$ is selected minimizing $\|\mathbf{u} - \mathbf{v}\|_{\Omega_e}$, the following inequality is verified

$$\frac{|R(\phi_h)|}{|\nabla \phi_h|} \leqslant \frac{|R(\psi_h)|}{|\nabla \psi_h|}, \quad \forall \psi_h \in V_h^p, \tag{20}$$

meaning that the CAU approximate solution ϕ_h is *optimal*. This is a remarkable feature of this method and shall be used in the linearization procedure. Moreover, combined with external adaptive refinement strategies this formulation locally searches for this minimum ratio accelerating convergence of the adaptive process (see [1,25]). In some sense the CAU method can also be seen as a self-adaptive consistent weighted residual method for which for a given $\phi_h \in V_h^p$ and the associated $R(\phi_h)$,

$$a(\phi_h, \eta_h) + \sum_{e=1}^{N_e} \big(R(\phi_h), p_c(\phi_h; \eta_h)\big)|_{\Omega_e} = (f, \eta_h), \quad \forall \eta_h \in V_h^p,$$

where the weighting perturbation function of the CAU formulation $p_c(\phi_h;\eta_h)$ depends on the searched approximate solution ϕ_h through $\mathbf{u}-\mathbf{v}(\phi_h)$ which also depends on the correspondent $R(\phi_h)$ and is defined as

$$p_c(\phi_h;\eta_h) = \left[\tau_s^p(\mathbf{u})\right]\mathbf{u}\cdot\nabla\eta_h + \left[\tau_c(\mathbf{u}-\mathbf{v})\right][\mathbf{u}-\mathbf{v}]\cdot\nabla\eta_h.$$

Introducing the nonlinear form

$$\begin{aligned} c(\phi_h;\phi_h,\eta_h)|_{\Omega_e} &= \big([\mathbf{u}-\mathbf{v}]\cdot\nabla\phi_h, \tau_c[\mathbf{u}-\mathbf{v}]\cdot\nabla\eta_h\big)|_{\Omega_e} \\ &= \big(\tau_c(\mathbf{u}-\mathbf{v})\otimes(\mathbf{u}-\mathbf{v})\nabla\phi_h, \nabla\eta_h\big)|_{\Omega_e} = (\tau_c\mathbf{C}\,\nabla\phi_h, \nabla\eta_h)|_{\Omega_e}, \end{aligned} \tag{21}$$

where the tensor $\mathbf{C} = (\mathbf{u}-\mathbf{v})\otimes(\mathbf{u}-\mathbf{v})$, the CAU method can be rewritten as: find $\phi_h \in V_h^p$, such that

$$D(\phi_h,\eta_h) + \sum_{e=1}^{N_e} c(\phi_h;\phi_h,\eta_h)|_{\Omega_e} = F(\eta_h), \quad \forall\eta_h \in U_h^p. \tag{22}$$

By design this formulation is consistent in the sense that, for a smooth solution ϕ satisfying (1), $R(\phi)=0$, $\mathbf{v}(\phi)=\mathbf{u}$, and

$$D(\phi,\eta_h) + \sum_{e=1}^{N_e} c(\phi;\phi,\eta_h)|_{\Omega_e} = F(\eta_h) \implies a(\phi,\eta_h) = (f,\eta_h), \quad \forall\eta_h \in U_h^p. \tag{23}$$

Moreover, it satisfies the error equation

$$D(e,\eta_h) - \sum_{e=1}^{N_e} c(\phi_h;\phi_h,\eta_h)|_{\Omega_e} = 0, \quad \forall\eta_h \in U_h^p, \tag{24}$$

which is a direct consequence of the consistency property (23). The tensor $\mathbf{C}$ in (21) has rank one and satisfies the eigenvalue problem $\mathbf{C}\psi = \lambda^2\psi$, whose solution is $\psi = (\mathbf{u}-\mathbf{v})/|\mathbf{u}-\mathbf{v}|$; $\lambda = |\mathbf{u}-\mathbf{v}| = |R(\phi_h)|/|\nabla\phi_h|$. Thus, $\mathbf{C}$ is a symmetric, positive semi-definite matrix. Thus, providing that $\tau_c \geqslant 0$, $c(\eta_h;\eta_h,\eta_h)|_{\Omega_e} \geqslant 0$ for any $\forall\eta_h \in U_h^p$, yielding the stability of the CAU method that follows immediately from the SUPG stability result,

$$D(\eta_h,\eta_h) + \sum_{e=1}^{N_e} c(\eta_h;\eta_h,\eta_h)|_{\Omega_e} \geqslant C(\theta)|||\eta_h|||^2. \tag{25}$$

For a general class of nonlinear shock-capturing methods, such as [20,16,11], the existence result is proven in [24]. As pointed by the authors, a local uniqueness result for the shock-capturing is still an open problem.

4.1. Iterative formulation

To solve the nonlinear equation (22) it is necessary to consider iterative methods which at each iteration preserve the main properties of the CAU method. This can be achieved using the following simple method: from the previous computed iterate solution ϕ_h^n, $n \in \mathcal{N}^+$, we get

$$c\big(\phi_h^n;\phi_h^{n+1},\eta_h\big)|_{\Omega_e} = \sum_{e=1}^{N_e}\big(R^n\big(\phi_h^{n+1}\big), \tau_c^n\big[\mathbf{u}-\mathbf{v}\big(\phi_h^n\big)\big]\cdot\nabla\eta_h\big)|_{\Omega_e}, \tag{26}$$

where

$$\tau_c^n = \tau_c\big(\mathbf{u} - \mathbf{v}\big(\phi_h^n\big)\big) \quad \text{and} \quad \big(\mathbf{u} - \mathbf{v}\big(\phi_h^n\big)\big) = \frac{R^n(\phi_h^n)}{|\nabla\phi_h^n|}\nabla\phi_h^n.$$

Hence, a simple iterative algorithm consists in: given ϕ_h^n and $[\mathbf{u} - \mathbf{v}(\phi_h^n)]$, find $\phi_h^{n+1} \in V_h^p$, such that

$$D\big(\phi_h^{n+1}, \eta_h\big) + \sum_{e=1}^{N_e} c\big(\phi_h^n; \phi_h^{n+1}, \eta_h\big)|_{\Omega_e} = F(\eta_h), \quad \forall \eta_h \in U_h^p, \tag{27}$$

for $\|\phi_h^{n+1} - \phi_h^n\|_\infty$ within a specified tolerance.

The next result states the convergence of the previous iterative algorithm.

Theorem. *Let the stability result* (25) *be valid. The sequence* $\{\phi_h^n\}$ *obtained with the iterative algorithm* (27) *converges to* ϕ_h *as* $n \to \infty$.

Proof. From (25) we get

$$\begin{aligned} C\big|\!\big|\!\big|\phi_h^{n+1}\big|\!\big|\!\big|^2 &\leqslant D\big(\phi_h^{n+1}, \phi_h^{n+1}\big) + \sum_{e=1}^{N_e} c\big(\phi_h^n; \phi_h^{n+1}, \phi_h^{n+1}\big)|_{\Omega_e} \\ &= \big(f, \phi_h^{n+1}\big) + \sum_{e=1}^{N_e}\big(f, \tau_s\mathbf{u}\cdot\nabla\phi_h^{n+1}\big)|_{\Omega_e} \leqslant \big|\big(f, \phi_h^{n+1}\big)\big| + \sum_{e=1}^{N_e}\big|\big(f, \tau_s\mathbf{u}\cdot\nabla\phi_h^{n+1}\big)|_{\Omega_e}\big|. \end{aligned}$$

Using the Young's inequality in the right side and the norm definition (10) yield

$$\begin{aligned} &2C\left(\varepsilon\big\|\nabla\phi_h^{n+1}\big\|^2 + \gamma\big\|\phi_h^{n+1}\big\|^2 + \sum_{e=1}^{N_e}\tau_s\big\|\mathbf{u}\cdot\nabla\phi_h^{n+1}\big\|_{\Omega_e}^2\right) \\ &\quad\leqslant C_1\gamma\big\|\phi_h^{n+1}\big\|^2 + C_2\sum_{e=1}^{N_e}\tau_s\big\|\mathbf{u}\cdot\nabla\phi_h^{n+1}\big\|_{\Omega_e}^2 + \big[(C_1\gamma)^{-1} + (C_2)^{-1}\big]\|f\|^2. \end{aligned}$$

Since C_1 and C_2 are arbitrary constants, they can be chosen appropriately to get

$$\big|\!\big|\!\big|\phi_h^{n+1}\big|\!\big|\!\big| \leqslant \overline{C}\|f\|,$$

showing that the sequence $\{\phi_h^n\}$ is bounded, which completes the proof. □

Remarks.

(1) The zero-th iterative solution ϕ_h^0 is the SUPG solution, which can be seen as the zero order residual correction for which $\tau_c^0 = 0$; $D(\phi_h^0, \eta_h) = F(\eta_h)$, $\forall \eta_h \in U_h^p$. The ϕ_h^0 would be the "optimal" CAU solution, provided $|R(\phi_h)|/|\nabla\phi_h|$ is independent of ϕ_h. This actually happens, for instance, for the steady one-dimensional hyperbolic problem $u\,\mathrm{d}\phi/\mathrm{d}x = 0$; $\phi_{|\Gamma_-} = g$, $\Gamma_- \subset \Gamma$ with $u \cdot n < 0$, for which $R(\phi_h) = u\,\mathrm{d}\phi_h/\mathrm{d}x$ and $|u - v(\phi_h)| = |u\frac{\mathrm{d}\phi_h}{\mathrm{d}x}|/|\frac{\mathrm{d}\phi_h}{\mathrm{d}x}| = |u|$. Consequently, the auxiliary velocity $v(\phi_h)$ vanishes yielding no residual correction in this case. Notice that this also occurs when the corresponding convective dominated diffusion problem,

$$u\frac{\mathrm{d}\phi}{\mathrm{d}x} - \varepsilon\frac{\mathrm{d}^2\phi}{\mathrm{d}x^2} = 0; \qquad \phi(0) = 0; \qquad \phi(1) = 1,$$

is approximated using linear interpolation. Under this situation, $\varepsilon \frac{\mathrm{d}^2\phi_h}{\mathrm{d}x^2} = 0$, and once again $v(\phi_h) = 0$ but a residual correction does exist for higher-order interpolations;

(2) Due to the property (20), any iterative solution satisfies $\frac{|R(\phi_h)|}{|\nabla\phi_h|} \leqslant \frac{|R(\phi_h^n)|}{|\nabla\phi_h^n|}$ which means that the convergence of the sequence $\{\phi_h^n\}$ to ϕ_h leads to

$$\frac{|R(\phi_h)|}{|\nabla\phi_h|} \leqslant \frac{|R(\phi_h^{n+1})|}{|\nabla\phi_h^{n+1}|} \leqslant \frac{|R(\phi_h^n)|}{|\nabla\phi_h^n|}. \tag{28}$$

4.2. Definition of the CAU upwind function

The stabilization parameter τ_c controls the local amount of additional stability provided by the term $c(\phi_h^n; \phi_h^{n+1}, \eta_h)|_{\Omega_e}$, which on the other hand depends on $(\mathbf{u} - \mathbf{v})$ in each element. Following the same directions used to design τ_s^p, the CAU stabilization parameter retains the functional dependence on $(\mathbf{u} - \mathbf{v}(\phi_h))$ in the following way: $\tau_c := h_c\xi_c/(2|\mathbf{u} - \mathbf{v}|p)$, where h_c and ξ_c are the characteristic element length and the non-dimensional numerical diffusivity associated with the $(\mathbf{u} - \mathbf{v})$ direction, respectively. To prevent over-diffusive approximate solutions, the CAU method should be designed such that it produces a minimum amount of dissipation along $\mathbf{u}$ since the SUPG term provides the control of derivatives along the streamline direction defined by the vector field $\mathbf{u}$. This requirement should be satisfied in some sense by the τ_c design. The dissipation due to the CAU term along the streamline direction amounts to

$$\varphi_c = \mathbf{C}\mathbf{u}\cdot\mathbf{u} = (\mathbf{u} - \mathbf{v}) \otimes (\mathbf{u} - \mathbf{v})\mathbf{u}\cdot\mathbf{u} = \big[(\mathbf{u} - \mathbf{v})\cdot\mathbf{u}\big]^2 = \big[|\mathbf{u}|^2 - \mathbf{u}\cdot\mathbf{v}\big]^2,$$

which shows that whenever $\mathbf{u}\cdot\mathbf{v} \leqslant 0$ the numerical dissipation introduced by the CAU is bigger than that provided by the SUPG ($\varphi_s = \mathbf{D}\mathbf{u}\cdot\mathbf{u} = (\mathbf{u}\otimes\mathbf{u})\mathbf{u}\cdot\mathbf{u} = |\mathbf{u}|^4$). This can be avoided by enforcing $\tau_c = 0$ when the approximate direction $\mathbf{v}(\phi_h)$ presents a component in the opposite direction to the true streamline $\mathbf{u}$. Hence, in each element Ω_e we select

$$\begin{cases} \tau_c = 0 & \text{when } \frac{|\mathbf{u}-\mathbf{v}|}{|\mathbf{u}|} \geqslant 1, \\ \tau_c = \tau_s^p\big[\frac{|\mathbf{u}|}{|\mathbf{u}-\mathbf{v}|} - 1\big] & \text{when } |\mathbf{u} - \mathbf{v}| < |\mathbf{u}|, \end{cases} \tag{29}$$

yielding

$$\begin{aligned} c\big(\phi_h^n; \phi_h^{n+1}, \eta_h\big)|_{\Omega_e} &= \left(\tau_c \frac{|R(\phi_h^n)|^2}{|\nabla\phi_h^n|^2}\nabla\phi_h^{n+1}, \nabla\eta_h\right)\Big|_{\Omega_e} \\ &= \left(\tau_s^p\left[\frac{|\mathbf{u}|}{|\mathbf{u} - \mathbf{v}|} - 1\right]|\mathbf{u} - \mathbf{v}|^2\nabla\phi_h^{n+1}, \nabla\eta_h\right)\Big|_{\Omega_e}. \end{aligned}$$

Hence, the local numerical diffusion introduced at each step by the shock-capturing term is $\bar{\varepsilon} = \tau_s^p[|\mathbf{u}||\mathbf{u} - \mathbf{v}| - |\mathbf{u} - \mathbf{v}|^2]$. In particular, its maximum value occurs whenever $|\mathbf{u} - \mathbf{v}| = |\mathbf{u}|/2$ yielding $\bar{\varepsilon}_{\max} = \tau_s|\mathbf{u}|^2/4$ and $\tau_c = \tau_s^p$. Thus, the maximum value of $\varphi_c = \mathbf{C}\mathbf{u}\cdot\mathbf{u} \leqslant |\mathbf{C}\mathbf{u}||\mathbf{u}| \leqslant \lambda^2|\mathbf{u}|^2 = |\mathbf{u} - \mathbf{v}|^2|\mathbf{u}|^2$ is $(\varphi_c)_{\max} \leqslant |\mathbf{u}|^4/4 \leqslant \varphi_s/4$ and $(\tau_c\varphi_c)_{\max} \leqslant (\varphi_s\tau_s^p)/4$. Regarding the definition (29), the maximum CAU dissipation will always be less than 25% of the SUPG dissipation. When $\mathbf{v}(\phi_h) = 0$, τ_c also vanishes, ensuring that no additional stability is added to the weak form.

A finer tuning on the τ_c definition can be obtained subtracting the SUPG contribution along the $(\mathbf{u}-\mathbf{v})$-direction, which is the direction of the approximate solution gradient. In this case τ_c is defined as (see [4]):

$$\tau_c = \tau_s^p \max\left(0, \left[\frac{|\mathbf{u}|}{|\mathbf{u}-\mathbf{v}|} - \alpha\right]\right), \quad \text{with } \alpha = \max\left(1, \frac{\mathbf{u}\cdot\nabla\phi_h}{R(\phi_h)}\right). \tag{30}$$

Notice that this expression reproduces (29) for the purely hyperbolic steady case. Also, for both definitions

$$\tau_c \leqslant \frac{|\mathbf{u}|}{|\mathbf{u}-\mathbf{v}|}\tau_s^p, \quad \text{with } \frac{|\mathbf{u}|}{|\mathbf{u}-\mathbf{v}|} > 1.$$

4.3. Error estimates

In this section we will prove that the additional term $c(\phi_h^n; \phi_h^{n+1}, \eta_h)$ does not degrade the rates of convergence as long as regular solutions are concerned. To this end, let $\tilde{\phi}_h \in V_h^p$ be the interpolant of ϕ and let us introduce the following definitions

$$e_h^{n+1} = \tilde{\phi}_h - \phi_h^{n+1}, \qquad \beta = \phi - \tilde{\phi}_h, \qquad e^{n+1} = \phi - \phi_h^{n+1} = e_h^{n+1} + \beta.$$

From the SUPG stability (9) and the CAU error equation (24), we obtain

$$\begin{aligned}
C\left|\!\left|\!\left|e_h^{n+1}\right|\!\right|\!\right|^2 &\leqslant D\left(e_h^{n+1}, e_h^{n+1}\right) = -D\left(\beta, e_h^{n+1}\right) + \sum_{e=1}^{N_e} c\left(\phi_h^n; \phi_h^{n+1}, e_h^{n+1}\right)|_{\Omega_e} \\
&\leqslant \left|\left(\mathbf{u}\cdot\nabla e_h^{n+1}, \beta\right)\right| + \left|\left(\varepsilon\nabla\beta, \nabla e_h^{n+1}\right)\right| + \left|\left(\gamma\beta, e_h^{n+1}\right)\right| \\
&\quad + \sum_{e=1}^{N_e}\left|\left(\mathbf{u}\cdot\nabla\beta, \tau_s^p\mathbf{u}\cdot\nabla e_h^{n+1}\right)|_{\Omega_e}\right| + \sum_{e=1}^{N_e}\left|\left(\varepsilon\Delta\beta, \tau_s^p\mathbf{u}\cdot\nabla e_h^{n+1}\right)|_{\Omega_e}\right| \\
&\quad + \sum_{e=1}^{N_e}\left|\left(\gamma\beta, \tau_s^p\mathbf{u}\cdot\nabla e_h^{n+1}\right)|_{\Omega_e}\right| + \sum_{e=1}^{N_e}\left|c\left(\phi_h^n; \phi_h^{n+1}, e_h^{n+1}\right)|_{\Omega_e}\right|,
\end{aligned}$$

where the incompressibility condition $\nabla\cdot\mathbf{u} = 0$ is used to write $(\mathbf{u}\cdot\nabla\beta, e_h^{n+1}) = -(\mathbf{u}\cdot\nabla e_h^{n+1}, \beta)$. The terms in the right-hand side are bounded in the following way using Young's inequality:

$$\begin{aligned}
&2\left(\mathbf{u}\cdot\nabla e_h^{n+1}, \beta\right) \leqslant C_1\tau_s^p\left\|\mathbf{u}\cdot\nabla e_h^{n+1}\right\|^2 + \left(C_1\tau_s^p\right)^{-1}\|\beta\|^2, \\
&2\left(\varepsilon\nabla\beta, \nabla e_h^{n+1}\right) \leqslant C_2\varepsilon\left\|\nabla e_h^{n+1}\right\|^2 + C_2^{-1}\left\|\sqrt{\varepsilon}\nabla\beta\right\|^2, \\
&2\left(\gamma\beta, e_h^{n+1}\right) \leqslant C_3\gamma\left\|e_h^{n+1}\right\|^2 + C_3^{-1}\left\|\sqrt{\gamma}\beta\right\|^2, \\
&2\left(\mathbf{u}\cdot\nabla\beta, \tau_s^p\mathbf{u}\cdot\nabla e_h^{n+1}\right)_{\Omega_e} \leqslant C_4\tau_s^p\left\|\mathbf{u}\cdot\nabla e_h^{n+1}\right\|_{\Omega_e}^2 + C_4^{-1}\left\|\sqrt{\tau_s^p}\mathbf{u}\cdot\nabla\beta\right\|_{\Omega_e}^2, \\
&2\left(\varepsilon\Delta\beta, \tau_s^p\mathbf{u}\cdot\nabla e_h^{n+1}\right)_{\Omega_e} \leqslant C_5\tau_s^p\left\|\mathbf{u}\cdot\nabla e_h^{n+1}\right\|_{\Omega_e}^2 + C_5^{-1}\left\|\sqrt{\tau_s^p}\varepsilon\Delta\beta\right\|_{\Omega_e}^2, \\
&2\left(\gamma\beta, \tau_s^p\mathbf{u}\cdot\nabla e_h^{n+1}\right)_{\Omega_e} \leqslant C_6\tau_s^p\left\|\mathbf{u}\cdot\nabla e_h^{n+1}\right\|_{\Omega_e}^2 + C_6^{-1}\left\|\sqrt{\tau_s^p}\gamma\beta\right\|_{\Omega_e}^2.
\end{aligned}$$

From (26) and (28) it is possible to bound the CAU term as

$$\begin{aligned}\left|c\left(\phi_h^n;\phi_h^{n+1},e_h^{n+1}\right)|_{\Omega_e}\right| &= \left|\left(R^n\left(\phi_h^{n+1}\right),\tau_c^n\left(\mathbf{u}-\mathbf{v}\left(\phi_h^n\right)\right)\cdot\nabla e_h^{n+1}\right)_{\Omega_e}\right| \\ &\leqslant \left(\left|R\left(\phi_h^{n+1}\right)\right|,\tau_c^n\left|\left(\mathbf{u}-\mathbf{v}\left(\phi_h^n\right)\right)\cdot\nabla e_h^{n+1}\right|\right)_{\Omega_e}.\end{aligned}$$

Besides, (29) and (30), imply

$$\tau_c^n\left|\left[\mathbf{u}-\mathbf{v}\left(\phi_h^n\right)\right]\cdot\nabla e_h^{n+1}\right|\leqslant \tau_s^p\left|\mathbf{u}\right|_\infty\left|\nabla e_h^{n+1}\right|,$$

yielding

$$\begin{aligned}2\left|c\left(\phi_h^n;\phi_h^{n+1},e_h^{n+1}\right)|_{\Omega_e}\right| &\leqslant 2\left(\left|R\left(\phi_h^{n+1}\right)\right|,\tau_s^p|\mathbf{u}|_\infty\left|\nabla e_h^{n+1}\right|\right)|_{\Omega_e} \\ &\leqslant C_7\varepsilon\left\|e_h^{n+1}\right\|_{\Omega_e}^2+C_7^{-1}\frac{|\mathbf{u}|_\infty^2(\tau_s^p)^2}{\varepsilon}\left\|R\left(\phi_h^{n+1}\right)\right\|_{\Omega_e}^2.\end{aligned}$$

Gathering the previous results and choosing the constants $C_1,\ldots,C_7$ accordingly we obtain

$$\begin{aligned}C\left|\left|\left|e_h^{n+1}\right|\right|\right|^2 &\leqslant \overline{C}\sum_{e=1}^{N_e}\frac{\tau_s^p}{\varepsilon}\left\|R\left(\phi_h^n\right)\right\|_{\Omega_e}^2 \\ &\quad + C'\sum_{e=1}^{N_e}\left(\varepsilon\|\nabla\beta\|_{\Omega_e}^2+\left(\tau_s^p\right)^{-1}\|\beta\|_{\Omega_e}^2+\tau_s^p\|\mathbf{u}\cdot\nabla\beta\|_{\Omega_e}^2+\varepsilon^2\tau_s^p\|\Delta\beta\|_{\Omega_e}^2\right).\end{aligned}$$

Assuming $\varepsilon=\mathrm{O}(h)$ and $\tau_s^p=\mathrm{O}(h/p)$, $\phi|_{\Omega_e}\in H^{k+1}(\Omega_e)$ for some $k\geqslant 1$, for any $0\leqslant s\leqslant\min(p,k)$, $p\geqslant 1$ and, in particular, if s is fixed and $p\gg 1$, using interpolation results we get

$$C''\left|\left|\left|e_h^{n+1}\right|\right|\right|^2\leqslant\overline{C}\sum_{e=1}^{N_e}\frac{h^2}{p^2\varepsilon}\left\|R\left(\phi_h^n\right)\right\|_{\Omega_e}^2+C\sum_{e=1}^{N_e}\left(\frac{h}{p}\right)^{2s+1}|\phi|_{s+1,\Omega_e}^2.$$

Considering that $|||e_h^{n+1}|||=|||e^{n+1}-\beta|||\geqslant|||e|||-|||\beta|||$ and $|||\beta|||\leqslant C\sum_{e=1}^{N_e}(\frac{h}{p})^{s+1/2}|\phi|_{s+1,\Omega_e}$, we get

$$\begin{aligned}\left|\left|\left|e^{n+1}\right|\right|\right| &\leqslant C\sum_{e=1}^{N_e}\left(\frac{h}{p}\right)^{s+1/2}|\phi|_{s+1,\Omega_e} \\ &\quad + C\sum_{e=1}^{N_e}\left[\frac{h^2}{p^2\varepsilon}\left\|R\left(\phi_h^n\right)\right\|_{\Omega_e}^2+\left(\frac{h}{p}\right)^{2s+1}|\phi|_{s+1,\Omega_e}^2\right]^{1/2}.\end{aligned}\tag{31}$$

Moreover,

$$\frac{h^2}{p^2\varepsilon}\left\|R\left(\phi_h^n\right)\right\|_{\Omega_e}^2\leqslant\widetilde{C}\left(\left(\frac{h}{p}\right)^2\varepsilon\left\|\Delta\left(\phi_h^n-\phi\right)\right\|_{\Omega_e}^2+\frac{h^2}{p^2\varepsilon}\left\|\mathbf{u}\cdot\nabla\left(\phi_h^n-\phi\right)\right\|_{\Omega_e}^2+\frac{h^2\gamma^2}{p^2\varepsilon}\left\|\phi_h^n-\phi\right\|_{\Omega_e}^2\right).$$

The initial guess of the iterative algorithm (when $n=0$) is the SUPG solution ϕ_h^0. Thus, considering the error estimate (15) for regular solutions and $\varepsilon=\mathrm{O}(h)$, we obtain

$$\frac{h}{p}\varepsilon^2\left\|\Delta\left(\phi_h^0-\phi\right)\right\|_{\Omega_e}^2+\frac{h}{p}\left\|\mathbf{u}\cdot\nabla\left(\phi_h^0-\phi\right)\right\|_{\Omega_e}^2+\gamma\left\|\left(\phi_h^0-\phi\right)\right\|_{\Omega_e}^2\leqslant C\left(\frac{h}{p}\right)^{2s+1}|\phi|_{s+1,\Omega_e}^2.$$

Grouping all the previous estimates for $n=0$, (31) yields:

$$\left|\left|\left|e^1\right|\right|\right|^2=\left|\left|\left|\phi_h^1-\phi\right|\right|\right|^2\leqslant C\left(\frac{h}{p}\right)^{2s+1}|\phi|_{s+1,\Omega_e}^2,$$

showing that for the first iteration the CAU method attains the same SUPG convergence rate owing to the choice (29) or (30). For $n > 0$ the same analysis applies and thus

$$\left|\left|\left|e^n\right|\right|\right|^2 \leqslant C \sum_{e=1}^{N_e} \left(\frac{h}{p}\right)^{2s+1} |\phi|^2_{s+1,\Omega_e}, \quad n = 1, 2, \ldots.$$

As the sequence $\{\phi_h^n\}$ converges to ϕ_h, we finally get

$$|||e|||^2 = |||\phi - \phi_h|||^2 \leqslant C \sum_{e=1}^{N_e} \left(\frac{h}{p}\right)^{2s+1} |\phi|^2_{s+1,\Omega_e},$$

which completes the proof.

5. Numerical examples

In this section we present numerical results obtained by applying the CAU method to the solution of three convection-dominated convection diffusion problems in which the medium is assumed homogeneous and isotropic, solved in a unit square domain $\Omega = (0, 1) \times (0, 1)$.

The first example deals with a problem with the source term f chosen so that the exact solution is the $C^\infty(\Omega)$ function

$$\phi(\mathbf{x}) = \sin(2\pi x),$$

with $\mathbf{u}^t = (1, 0)$, $\varepsilon = 10^{-8}$ and an inflow $\phi = 0$. The error in the solution, measured in the triple norm (10), is obtained with varying h or p. For uniform refinements of a triangular mesh, the results are listed in Table 1. Table 2 shows the error in the solution obtained for quasiuniform h-refinement of the mesh depicted in Fig. 1. In Table 2, the characteristic lenght h is the greatest element lenght in the mesh.

We first consider the case when the interpolation order p is fixed and the characteristic lenght h is varied as shown in Fig. 2. The slope of the lines corresponding to a fixed value of p verifies the expected $p + 1/2$ convergence rate for both the uniform and quasiuniform meshes. Fig. 3 shows the error in the solution when h is fixed and p is varied. The slope of the lines which increases as p increases verifies the expected exponential rates of convergence owing to the regularity of the exact function. In Fig. 4 we combine the previous results showing the error in the solution as a function of the total number of unknowns. This figure highlights the fact that an improvement in accuracy is obtained by using higher interpolation orders for the same number of unknowns regarding smooth solutions.

Table 1
Example 1—Error with uniform h and p

mesh	$-\log h$	$-\log \lvert\lvert\lvert\phi - \phi_h\rvert\rvert\rvert$			
		$p = 1$	$p = 2$	$p = 3$	$p = 4$
4×4	0.301	0.0052	0.9136	1.8844	2.9650
8×8	0.602	0.4622	1.6383	2.9234	4.3002
16×16	0.903	0.9270	2.3732	3.9719	5.6472
32×32	1.204	1.5181	3.2518	5.2616	7.3977

Table 2
Example 1—Error with quasiuniform h and p

mesh	$-\log h$	$-\log \lVert\lVert \phi - \phi_h \rVert\rVert$			
		$p=1$	$p=2$	$p=3$	$p=4$
4×4	0.523	−0.0399	0.8107	1.7405	2.7041
8×8	0.824	0.4140	1.5259	2.7679	4.0297
16×16	1.125	0.8800	2.2575	3.8125	5.3707
32×32	1.426	1.3381	3.0010	4.8629	6.7191

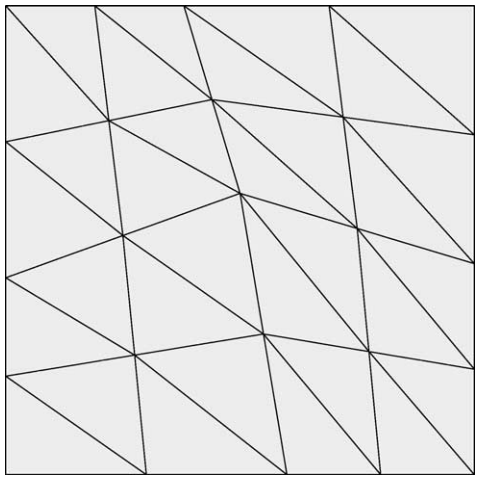

Fig. 1. Example 1: Quasiuniform mesh.

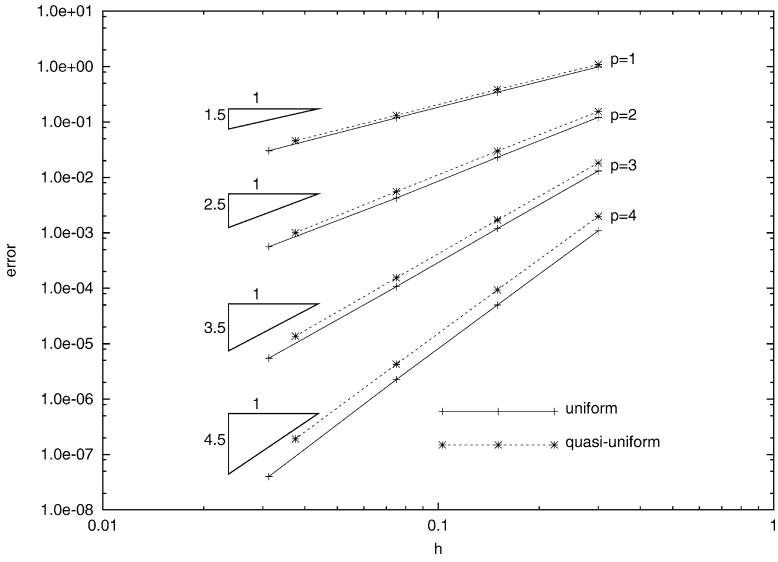

Fig. 2. Example 1: Convergence rates for the CAU (fixed p).

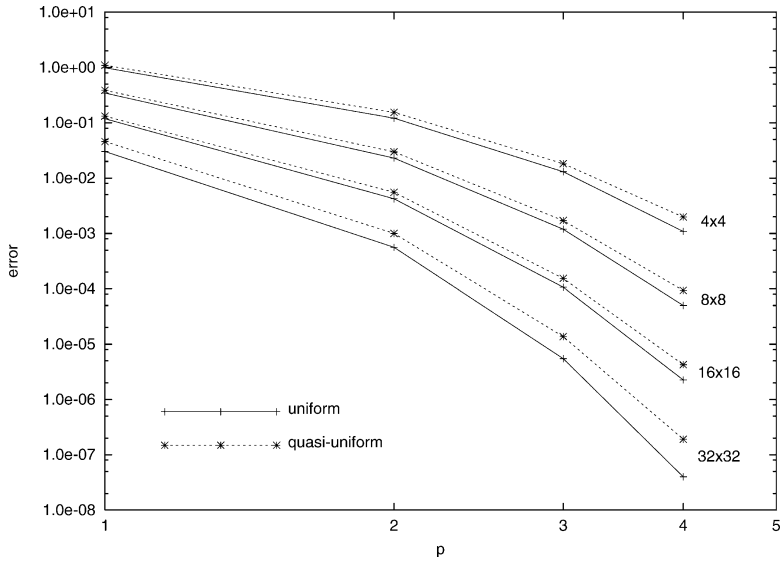

Fig. 3. Example 1: Convergence rates for the CAU (fixed h).

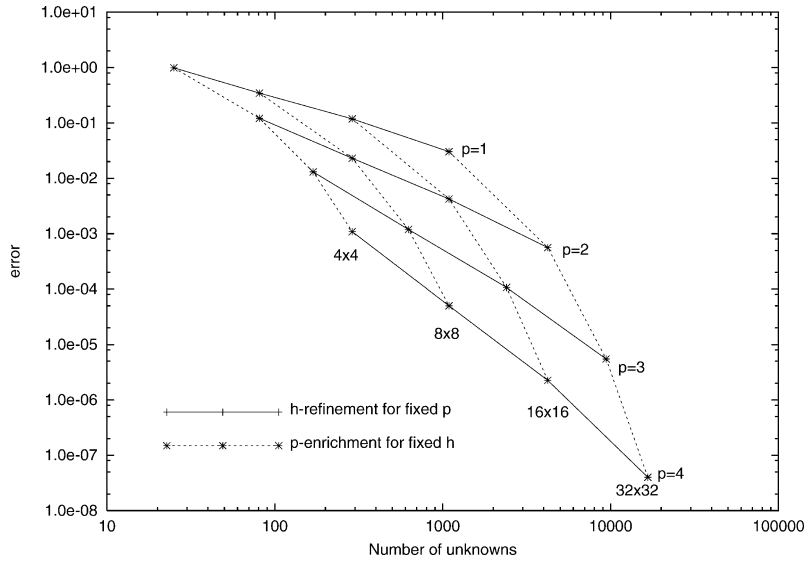

Fig. 4. Example 1: Convergence rates for the CAU.

The second example deals with a two-dimensional problem with a source term such that the exact solution is the continuous function

$$\phi(\mathbf{x}) = \sin(\pi x)\sin(\pi y),$$

with uniform 45° skew velocity field flow with $|\mathbf{u}| = 1$ and homogeneous Dirichlet boundary conditions. Fig. 5 shows the convergence rates for $\varepsilon = 10^{-8}$ for the same uniform meshes used previously. The slope of the lines corresponding to a fixed value of the interpolation order p again verifies the expected

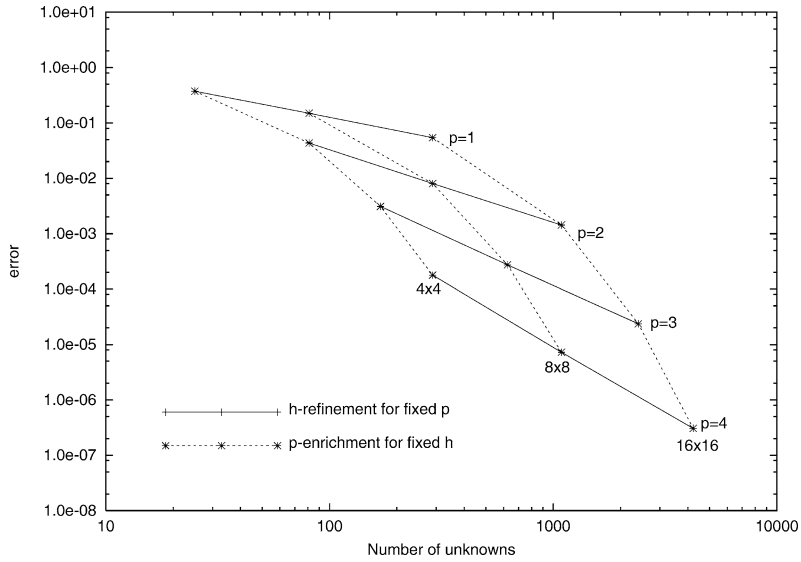

Fig. 5. Example 2: Convergence rates for the CAU ($\varepsilon = 10^{-8}$).

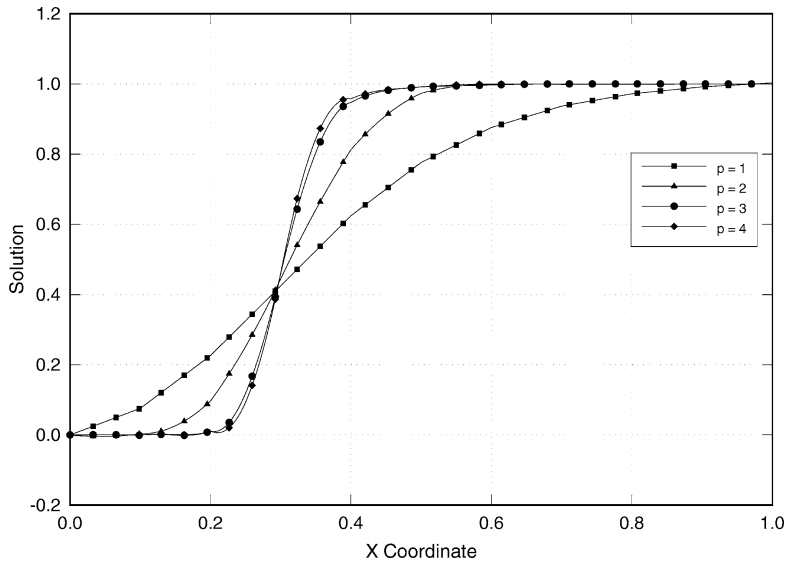

Fig. 6. Example 3: CAU approximate solution at $y = 0.5$.

$p + 1/2$ rate. We also evaluate the error in the solution obtained for $\varepsilon = 10^{-1}$ resulting in the same behavior obtained for $\varepsilon = 10^{-8}$. For this reason, it is not depicted here.

Finally, the last numerical example is the well-known problem of the advection skewed with respect to the mesh. The source term is defined as $f(\mathbf{x}) = 0$, the diffusivity coefficient is $\varepsilon = 10^{-8}$, the velocity field is $\mathbf{u}^t = (1, 1)$ with the following Dirichlet boundary conditions: $\phi(x, 0) = 1$; $\phi(0, y) = 1$ if $y \leqslant 0.2$ and $\phi(0, y) = 0$ if $y > 0.2$. A (sharp) inner boundary layer along the streamline is formed in the inflow, caused by the change of the Dirichlet boundary condition. We consider the case when the characteristic

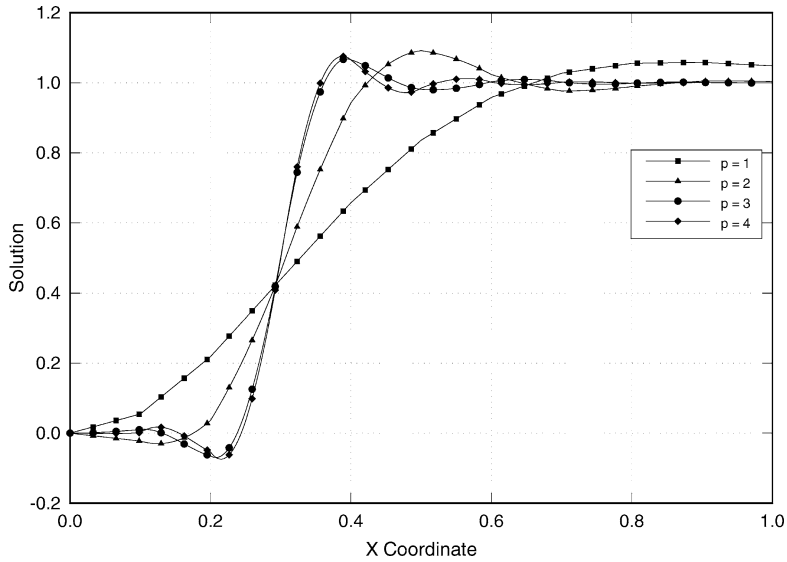

Fig. 7. Example 3: SUPG approximate solution at $y = 0.5$.

lenght h is fixed, corresponding to a uniformly divided unit square with 10 divisions, and p is varied. The numerical solution at the mid-section $y = 0.5$ obtained with a uniform p-enrichment is shown in Figs. 6 and 7 for the CAU and SUPG methods, respectively. It is clear that an improvement in accuracy is obtained for the CAU method by using higher-order elements while keeping stability. This feature can be successfully used to save computational cost in deriving accurate numerical solutions of convection dominated problems with regions containing discontinuities or steep gradients [1].

6. Conclusions

In this paper the CAU method and its version for higher interpolation orders are analysed. It is shown that the CAU method preserves the same SUPG convergence rates for regular solutions owing to the appropriate design of the stabilization parameters proposed here. Such design also guarantees the same behavior at the diffusive limit. Those estimates are ratified by one- and two-dimensional numerical experiments, considering finite elements interpolation orders up to the fourth degree. The good stability properties of the CAU in representing steep gradients as well as its accuracy turn it a useful tool to solve problems in computational fluid dynamics when convection phenomena are dominant.

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